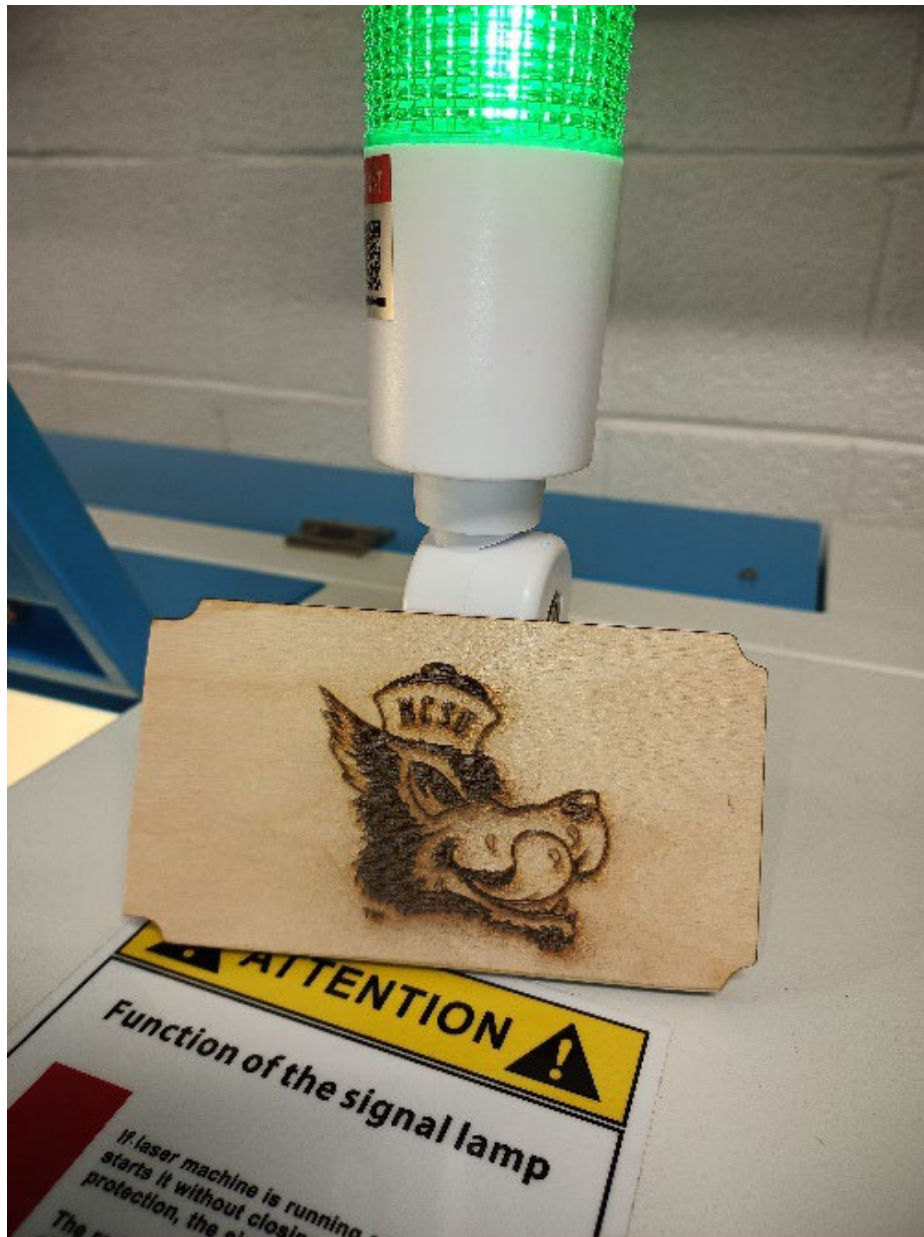


Thunder Laser System Nova 35 80 W HOW-TO GUIDE

Sean Ryan, Justin Egresitz, Joseph Kaskel, Dr. Steven Miller



Plug in the laser cutter:

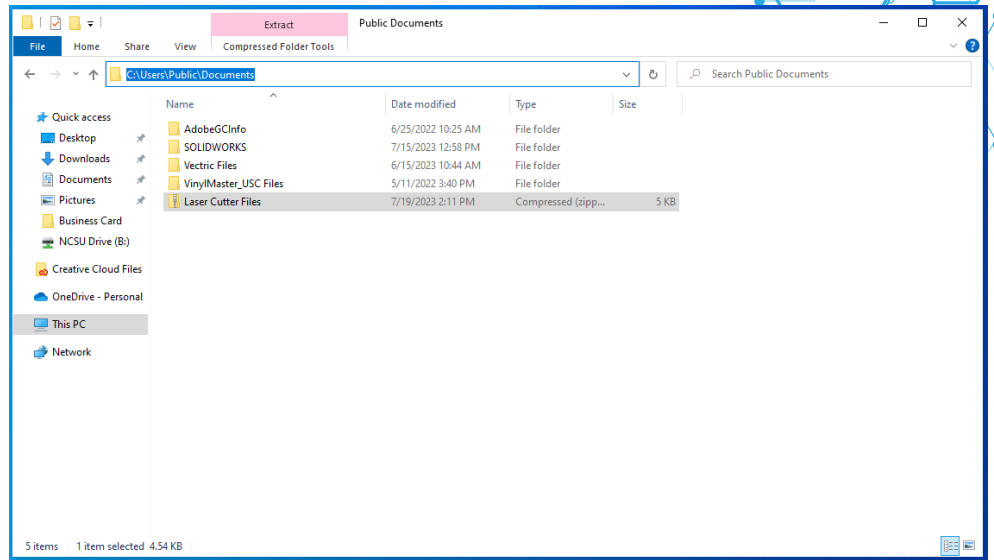


In the following order:
1. Turn on the Main switch
2. Turn on the Laser switch



Prep your files. Go to the public folder (C:\Users\Public\Documents) on the Poe Laser Cutter Computer.

COPY AND PASTE THE "Laser Cutter Files" ZIP TO YOUR DOCUMENTS. DO NOT MODIFY THE PUBLIC DOCUMENTS



 Laser Setup.lbdev 

 Thunder_80_Watt_LightBurn_Materials_Library.clb 

 ThunderGeneralTemplate.lbrn2 

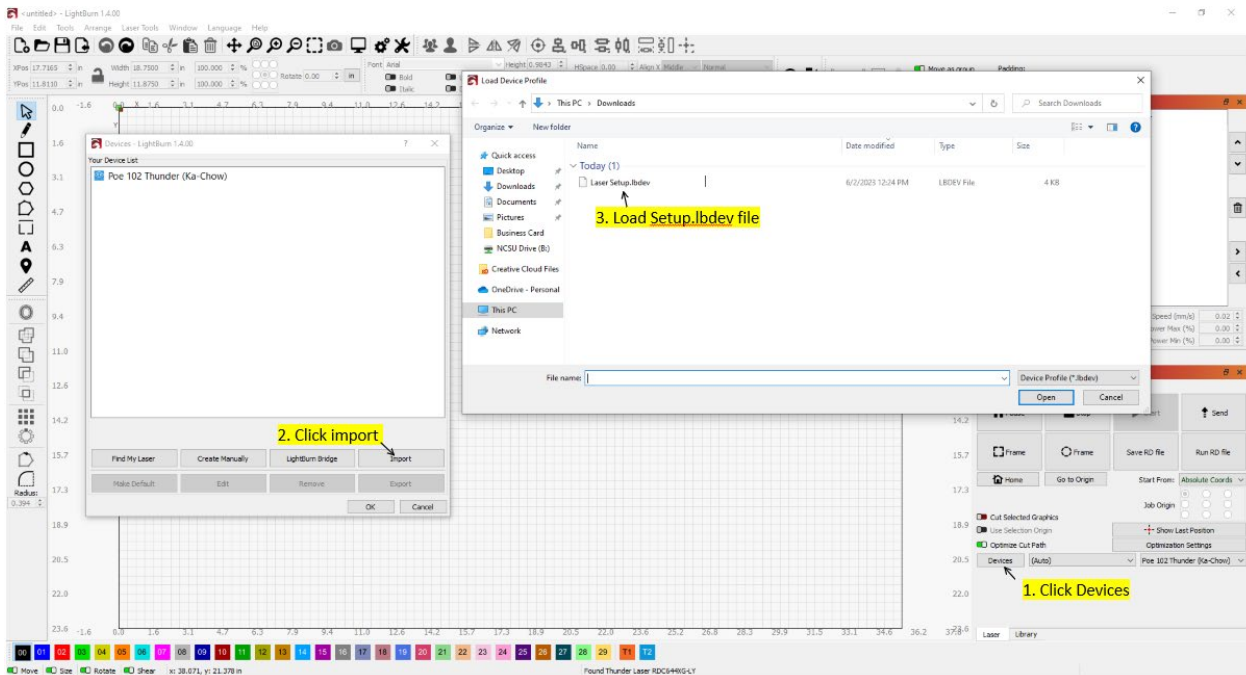
Laser Setup.lbdev is a config file based on manufacturer recommendations. You won't need to complete a full install if using the Poe lab computer. However, you will need to import the Laser Setup.lbdev config file.

If you do need to do an entire install, use the cheat sheet provided on the manufacturer's website.

https://support.thunderlaserusa.com/portal/en/kb/articles/cheat-sheet-important-preparation-first-steps#Enter_Scanning_Offsets_from_Gold_Card

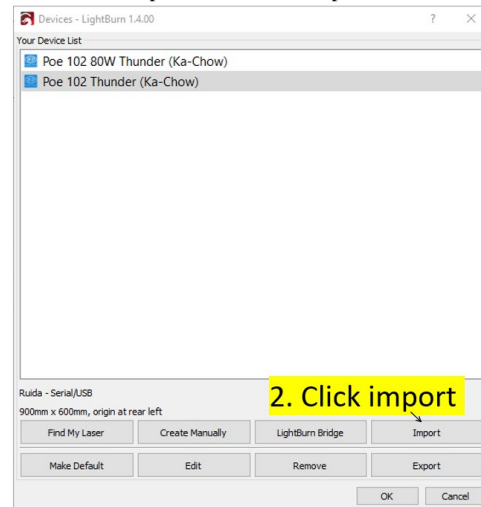
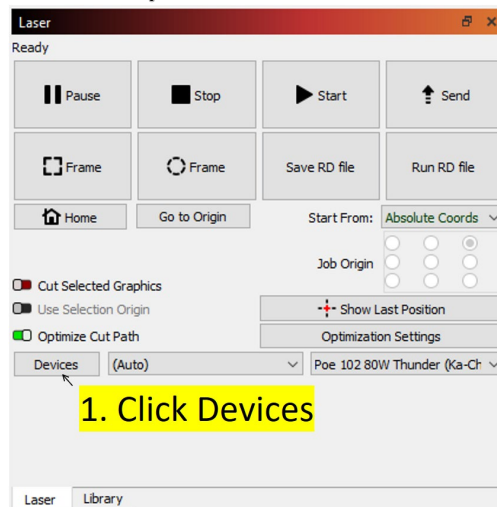
 Laser Setup.lbdev 

After starting LightBurn, load the Setup.lbdev file.

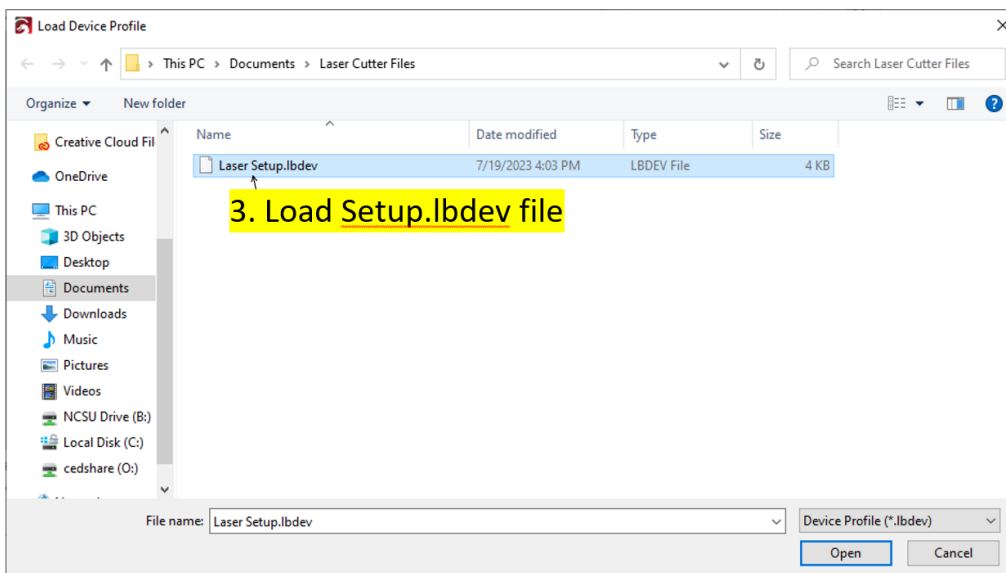


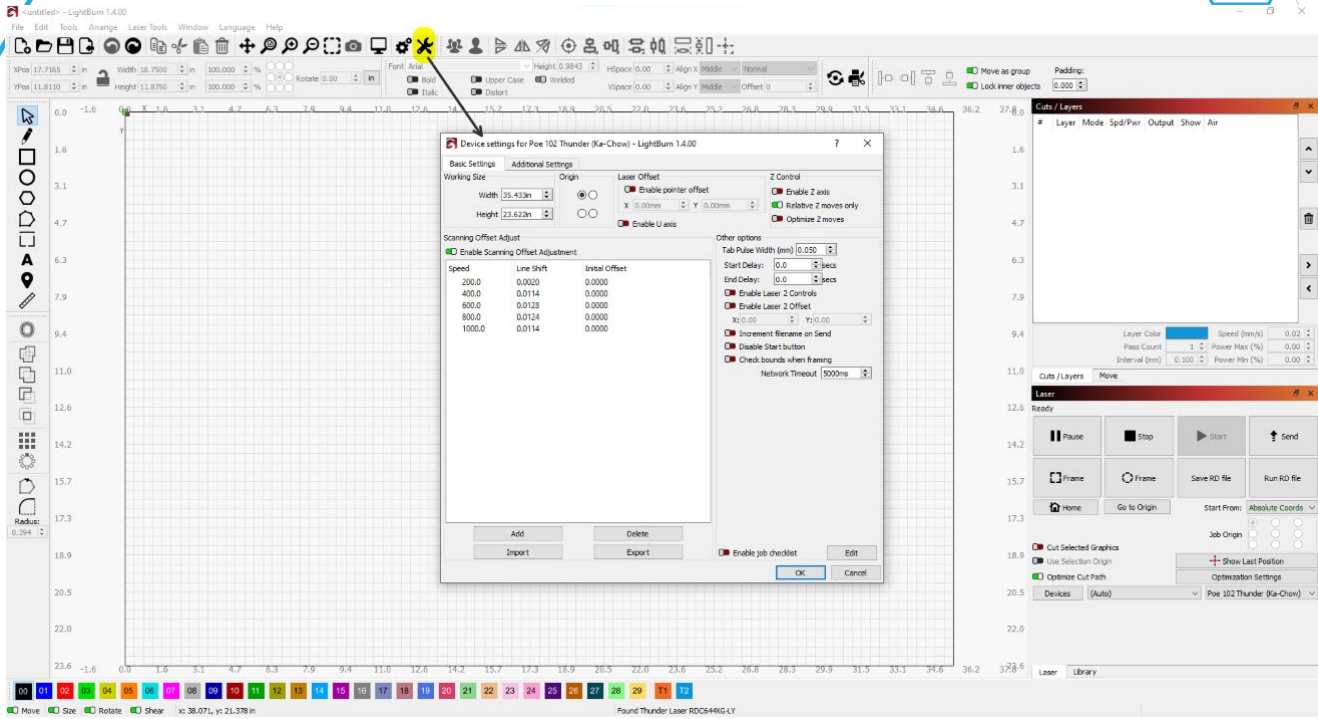
Step 1. Click Devices

Step 2. Click Import

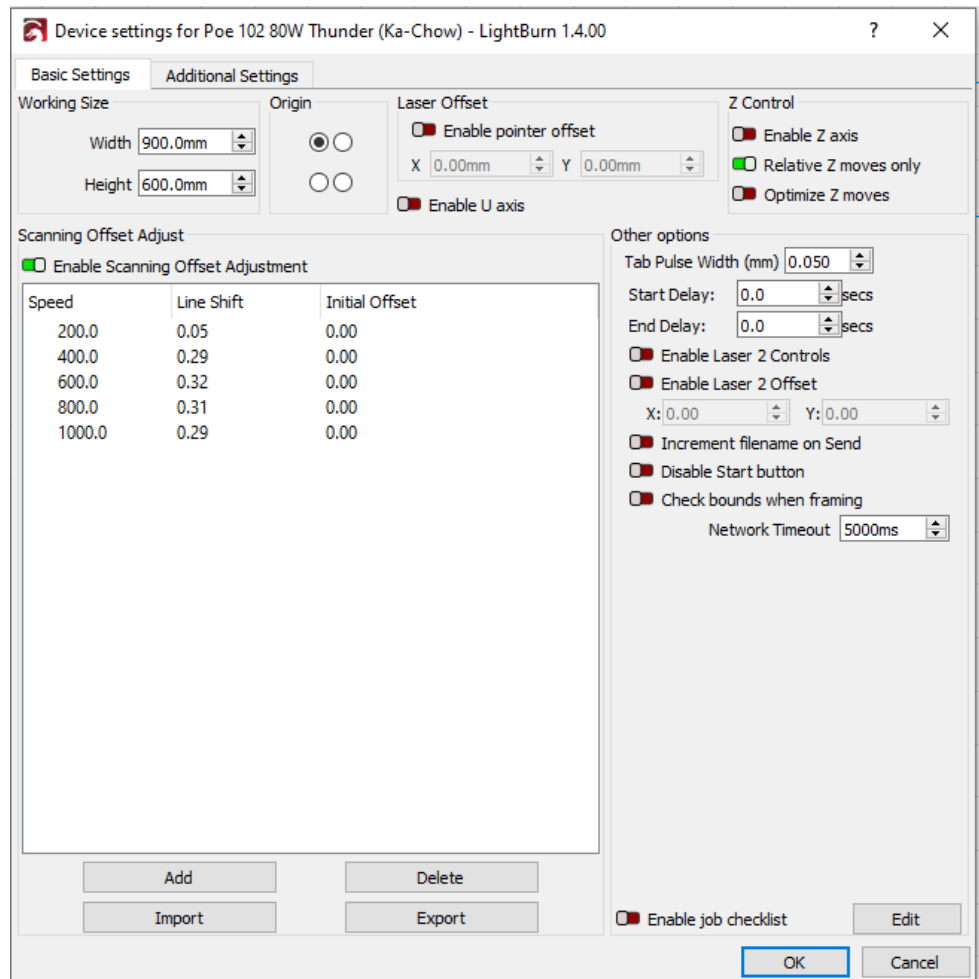


Step 3. Open the Laser Setup.lbdev file.





To ensure this was loaded correctly, click Device Settings. These should match the manufacturer's recommended gold card settings.



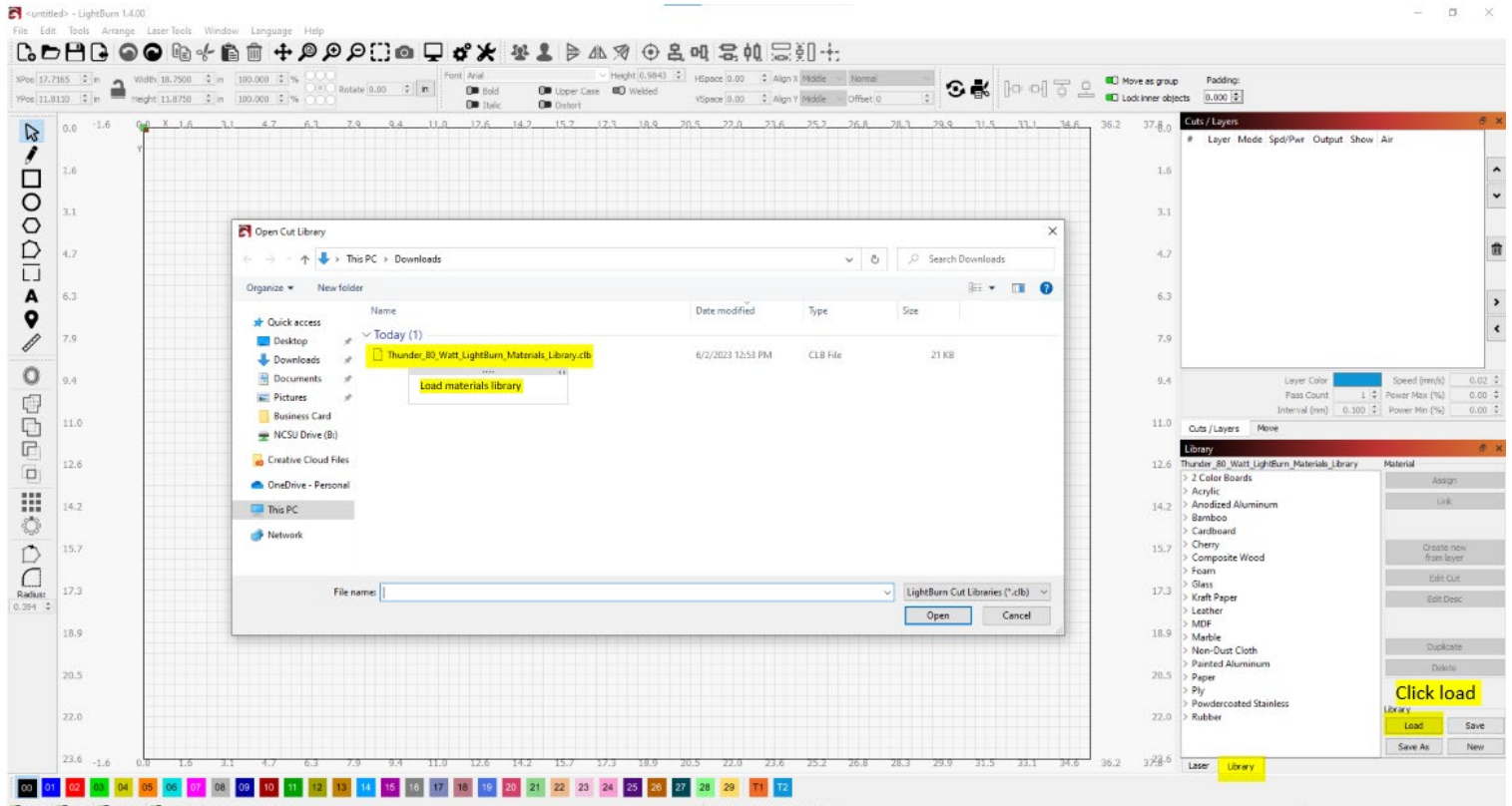
Next, you'll want to load the materials library provided by the manufacturer.

The file can be found in Poe's Laser Cut Folder.

It also can be found at the link below: <https://support.thunderlaserusa.com/portal/en/kb/articles/80-watt-materials-libraries>

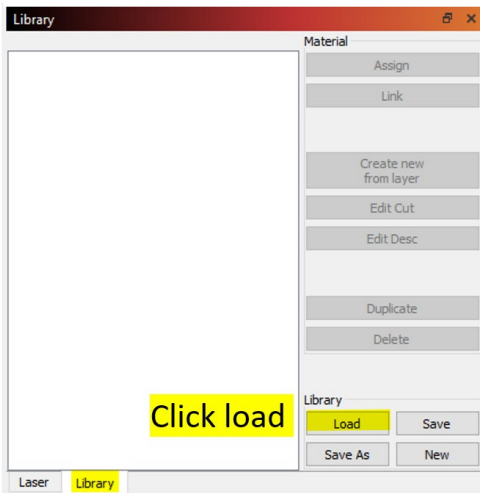


Thunder_80_Watt_LightBurn_Materials_Library.clb



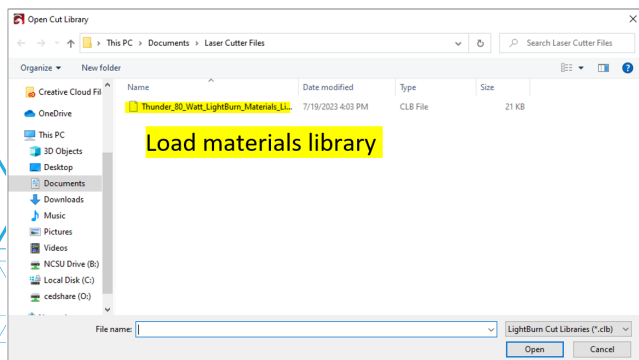
Step 1. Click the Library tab and then the Load button.

Step 3. You should now see materials as shown in the picture below.

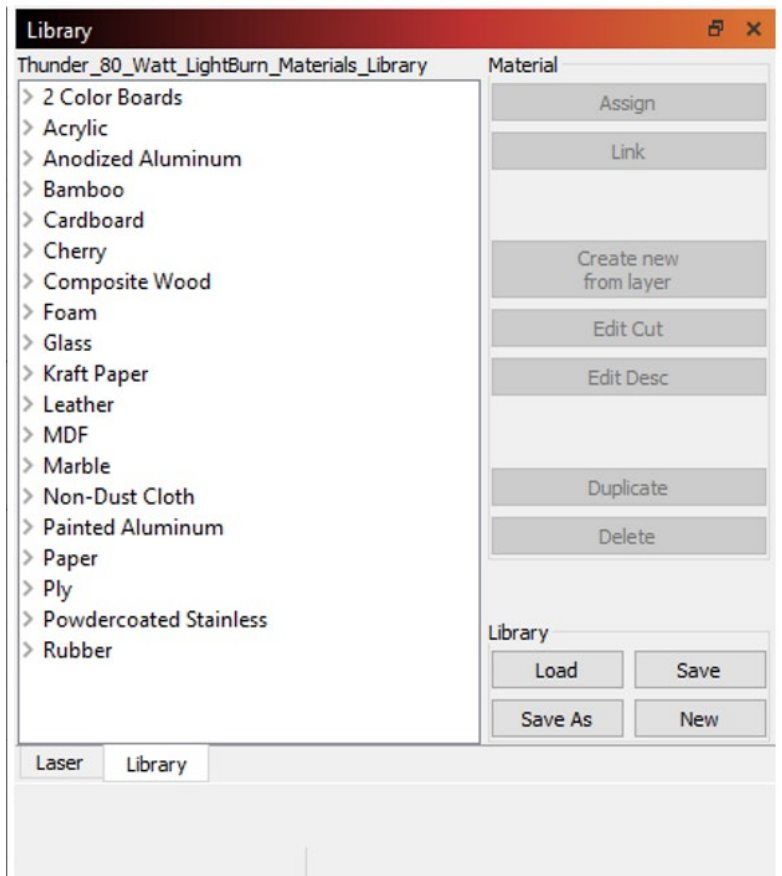


Click load

Step 2. Load the materials library from Poe's Laser Cutter Files.



Load materials library



The following steps are workflow recommendations and standards.

Open the

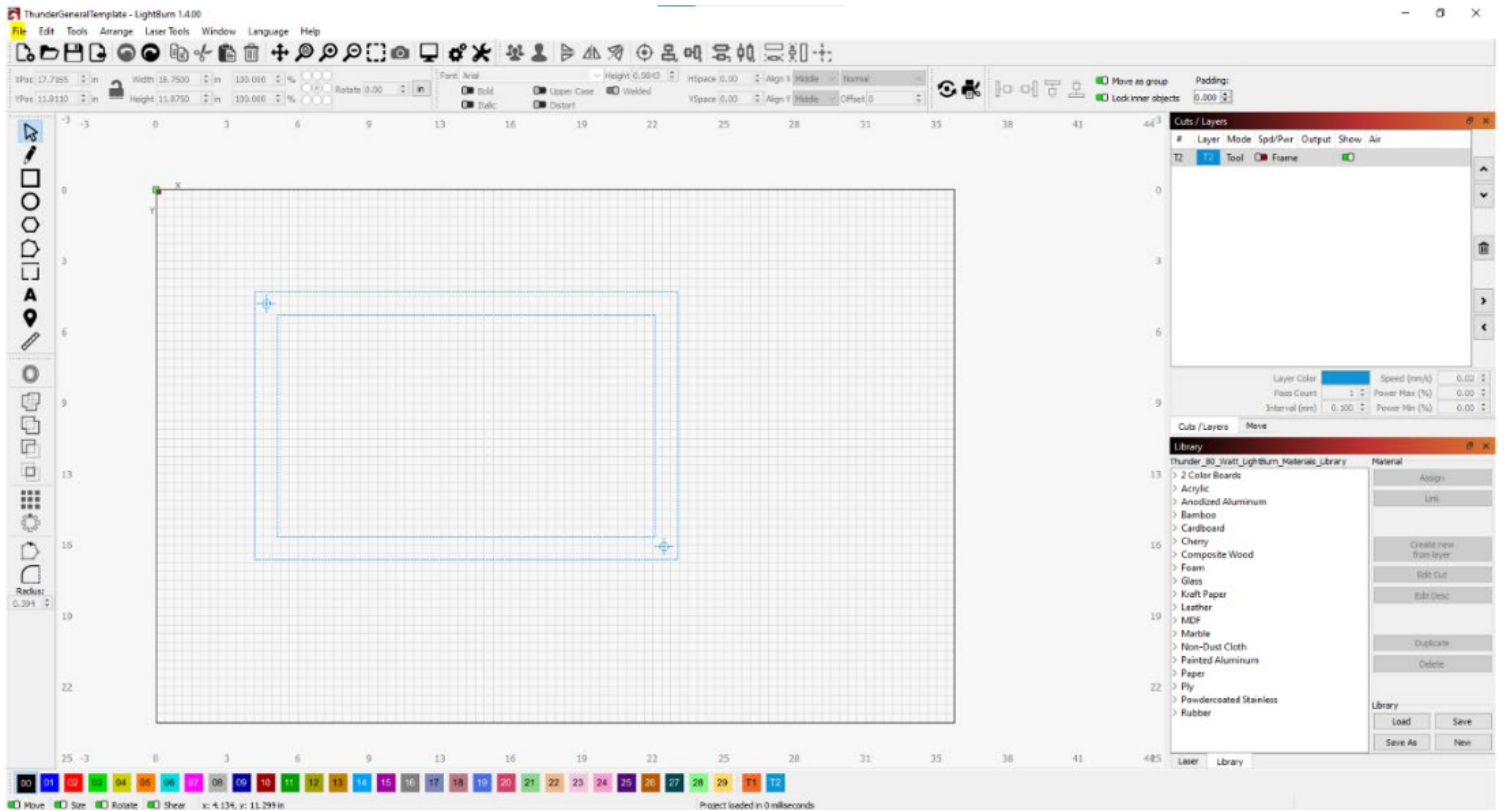
"ThunderGeneralTemplate.lbrn2" file



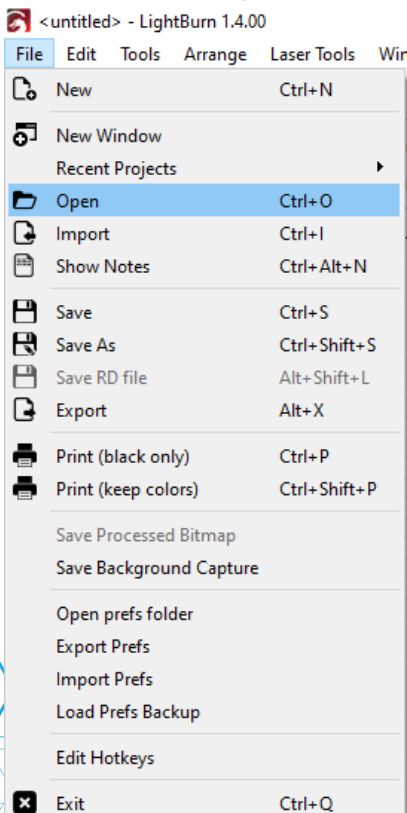
ThunderGeneralTemplate.lbrn2



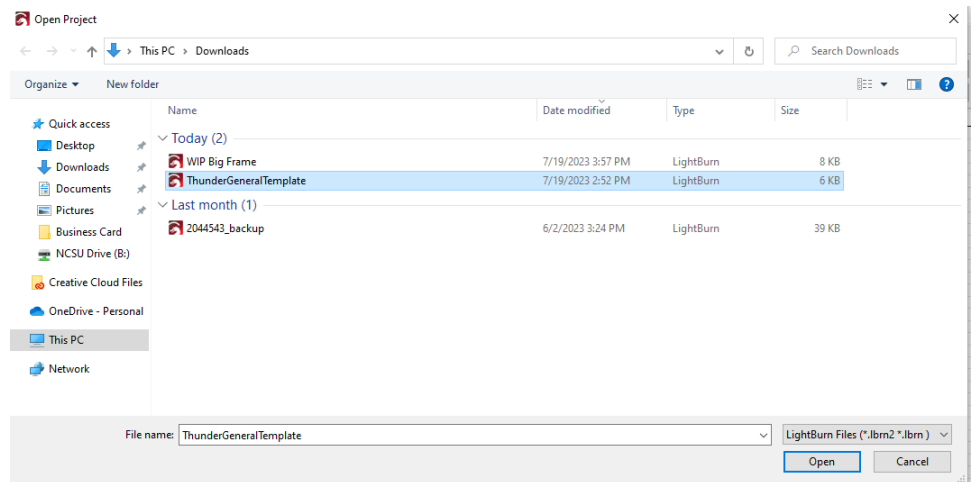
1. Click on "File" in the left-hand corner.



2. Click "Open"



3. Click "ThunderGeneralTemplate"



Note that this blue box will be your workspace.

NCSU has multiple templates in varied sizes to work with different projects.

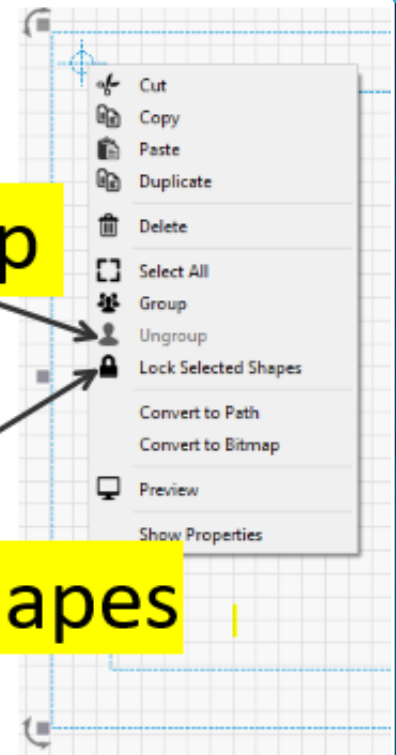
Highlight your template and right-click to ungroup shapes and focus on the cross mark.

1. Right-click your template and unlock the selected shapes.

2. Right-click your template and click Ungroup Selected Shapes.

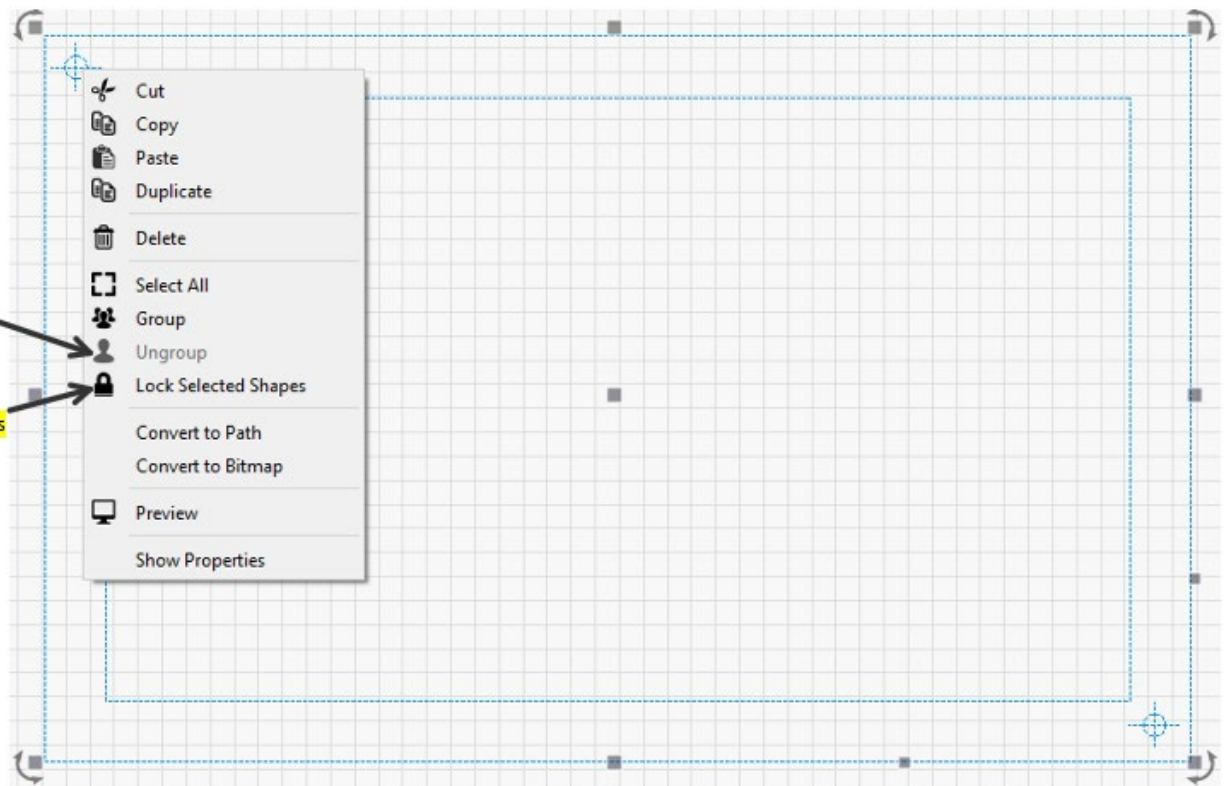
2. Ungroup

1. Unlock selected shapes



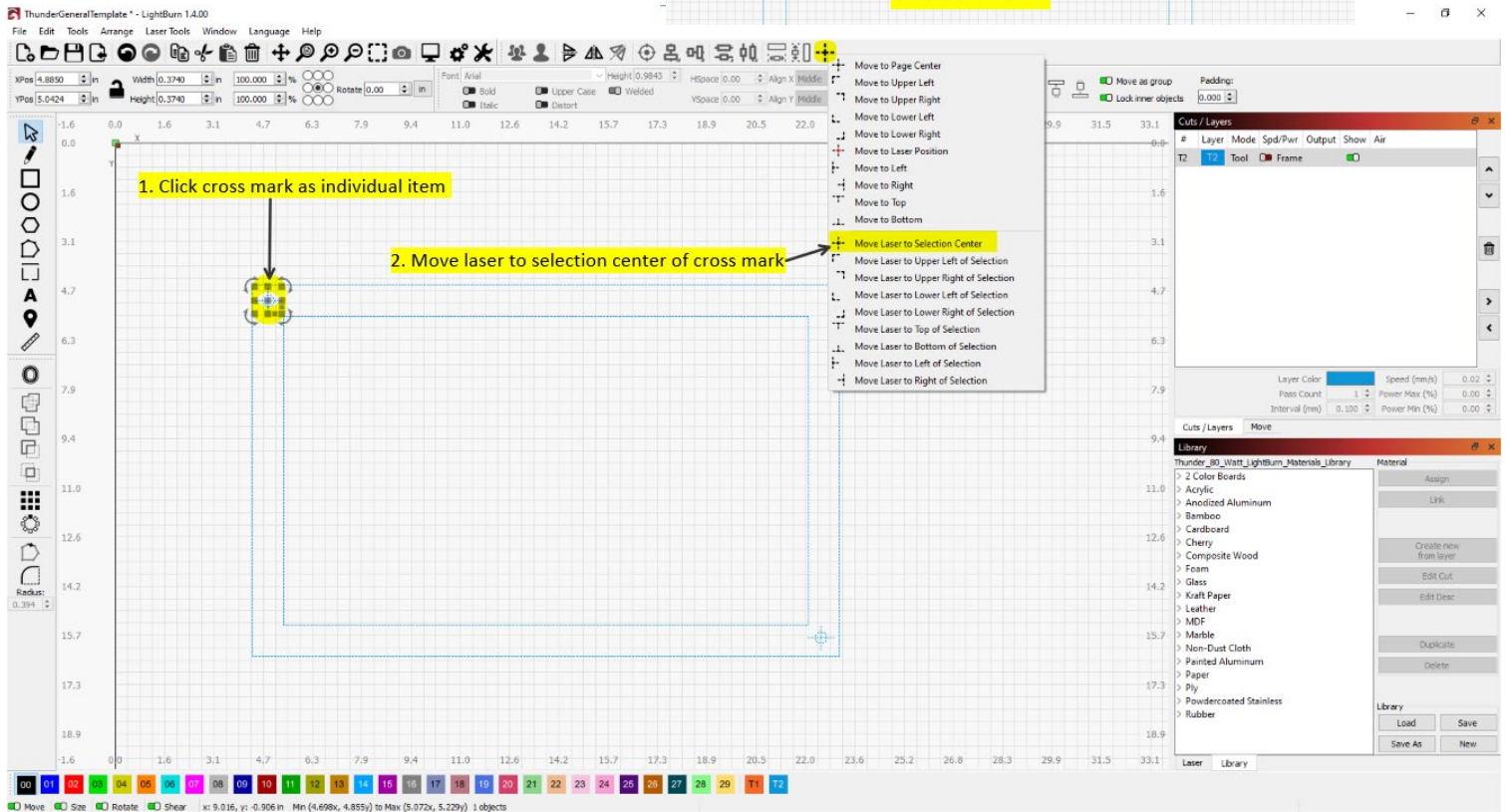
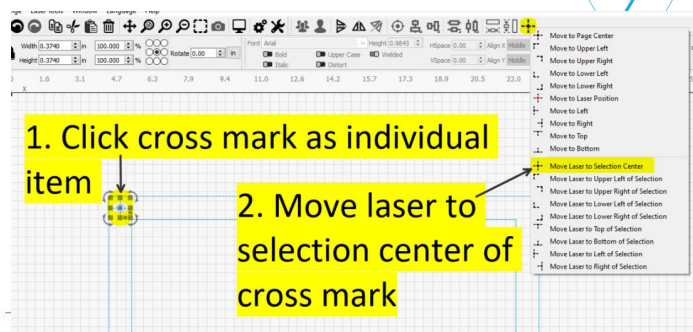
2. Ungroup

1. Unlock selected shapes



1. Right-click the cross mark as an individual item.

2. Click on "Move laser to selection center".

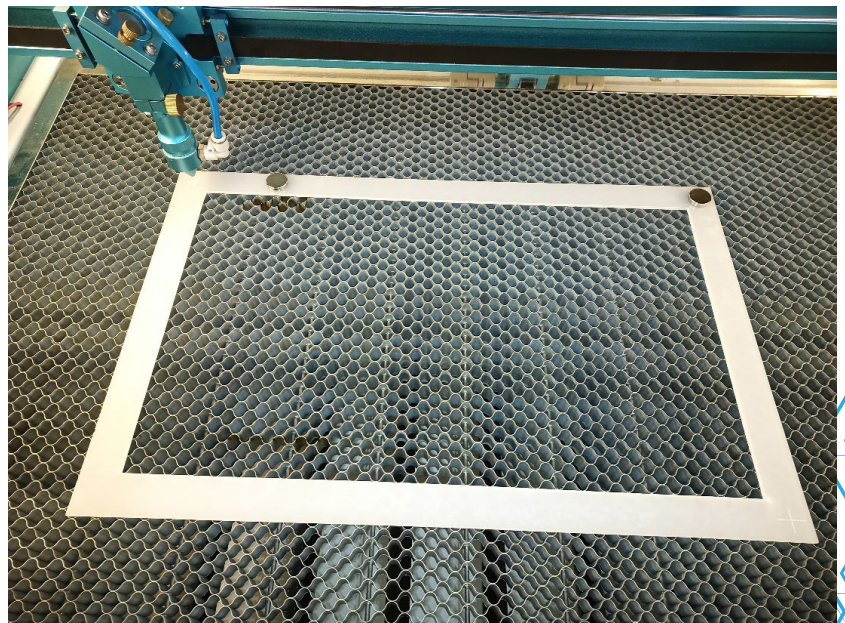
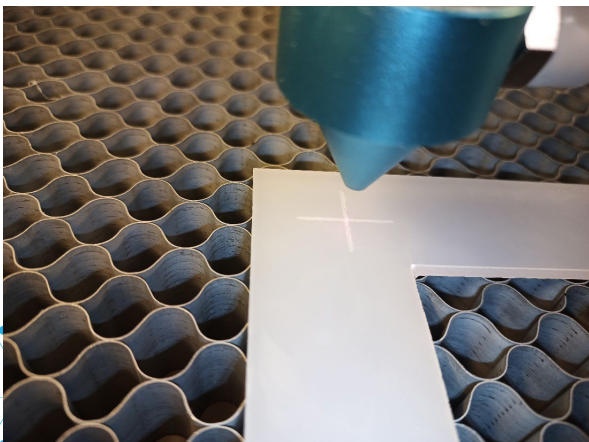


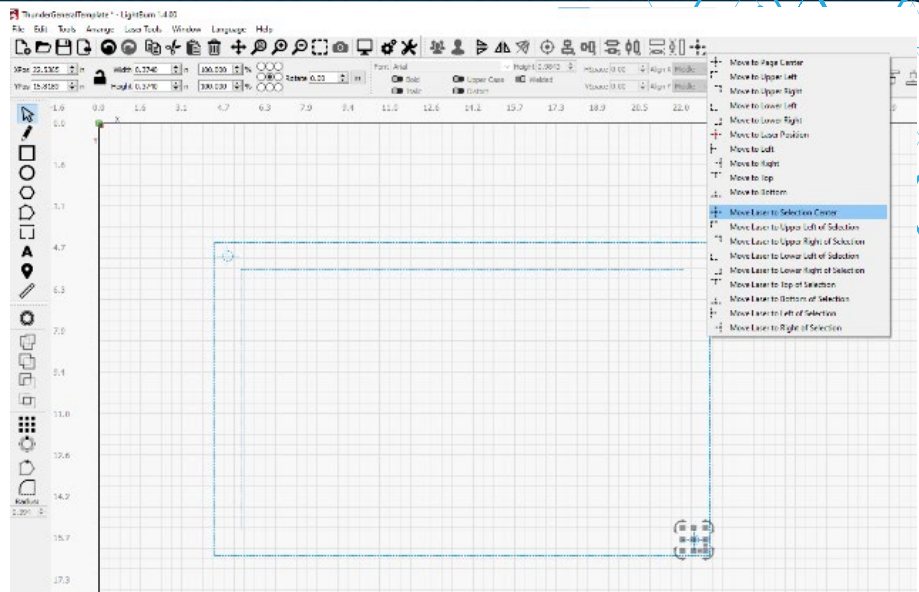
You want your template a few inches away from the end walls for the best performance, as the motor will have trouble rastering correctly if it is too close to the edge.

In real life, take your template and line up the cross mark with the laser dot.

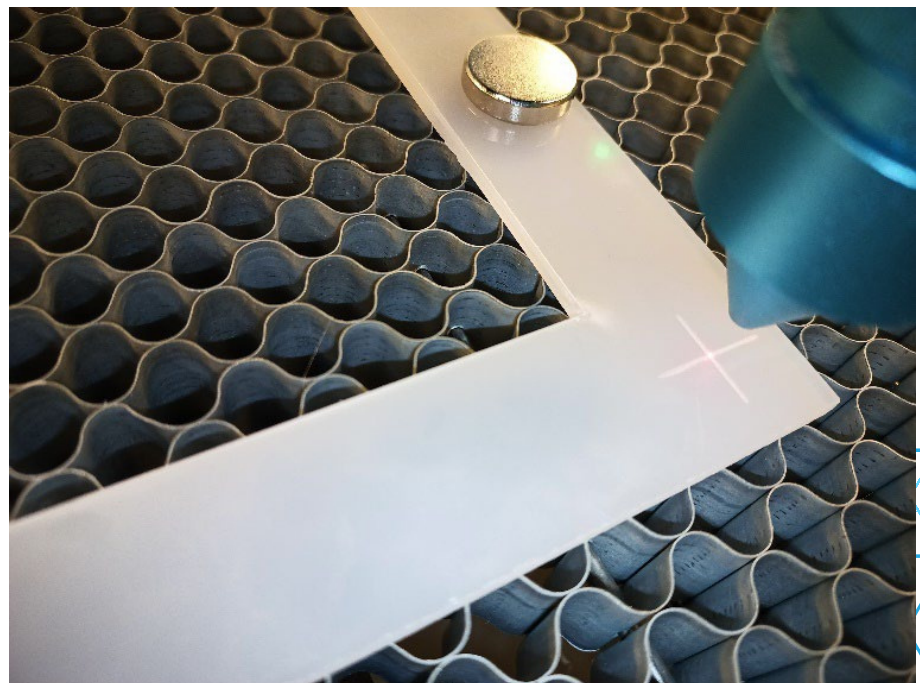
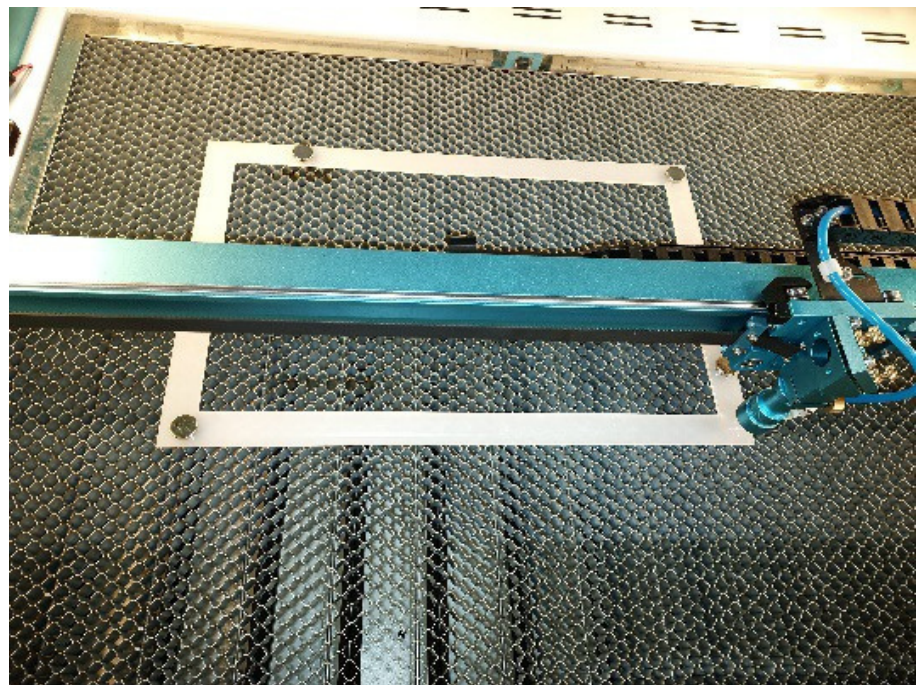
NOTE: Do NOT just place the template into the corner closest to the edge.

TIP: Utilize magnets to hold your template in place.

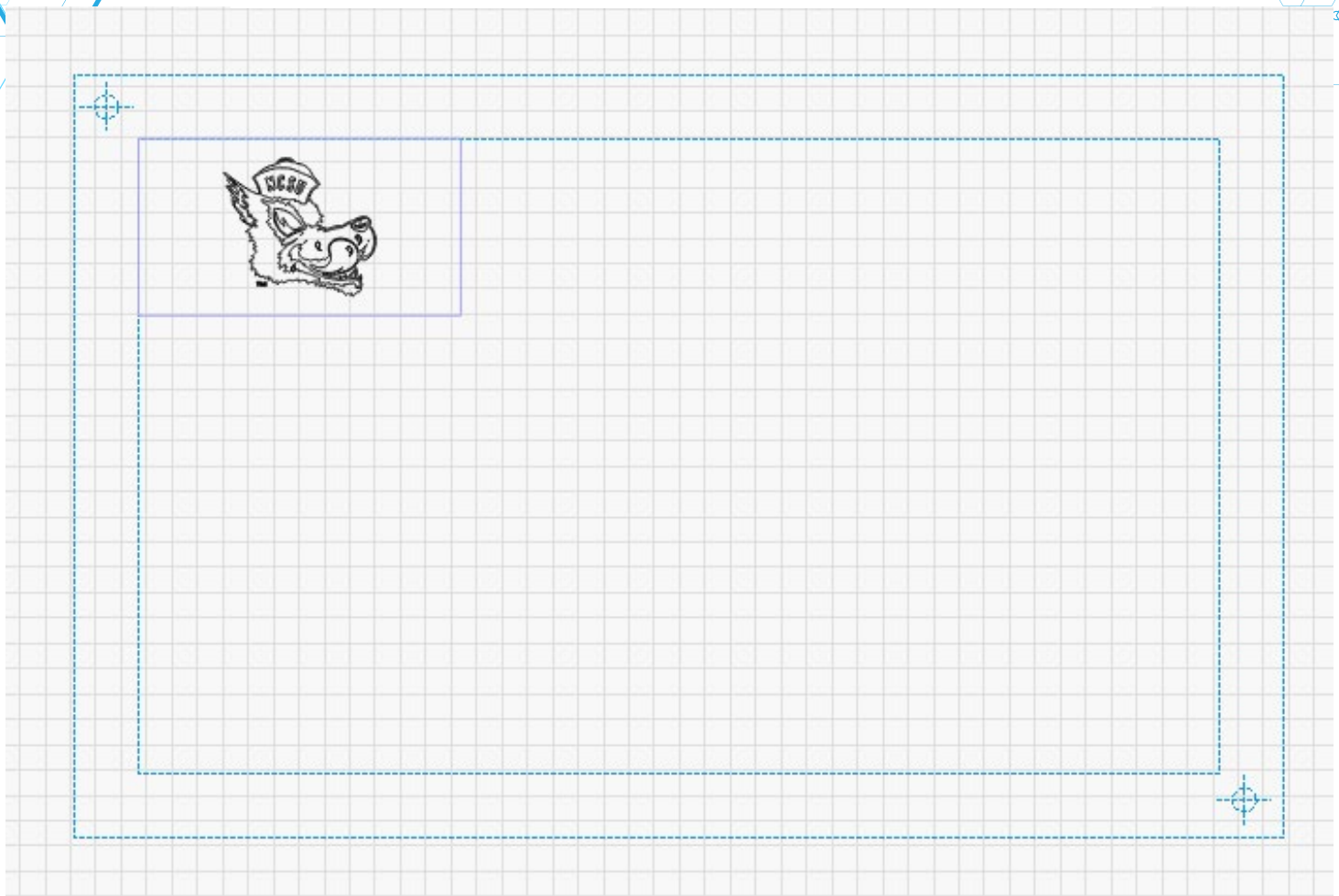




Perform the same steps on the cross mark in the opposite corner. This is to ensure your template is straight.



You are now ready to place your project.
Open your SVG from the File menu in the upper-left corner.



Turning off output will have the laser ignore everything associated with that layer.

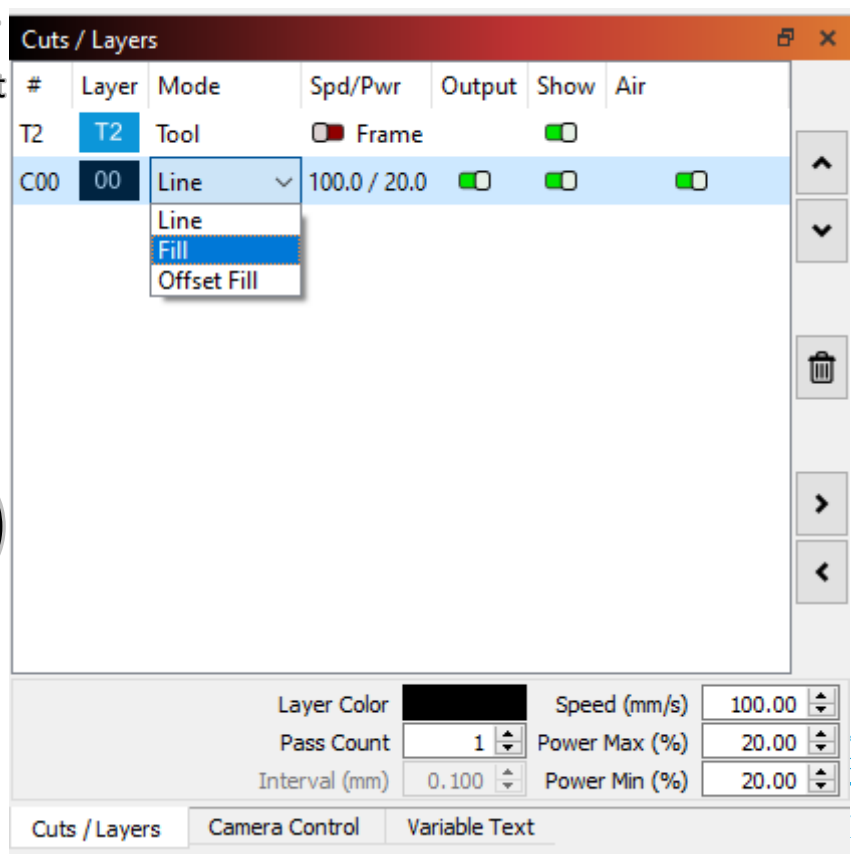
The tool mode (template) will also be ignored.

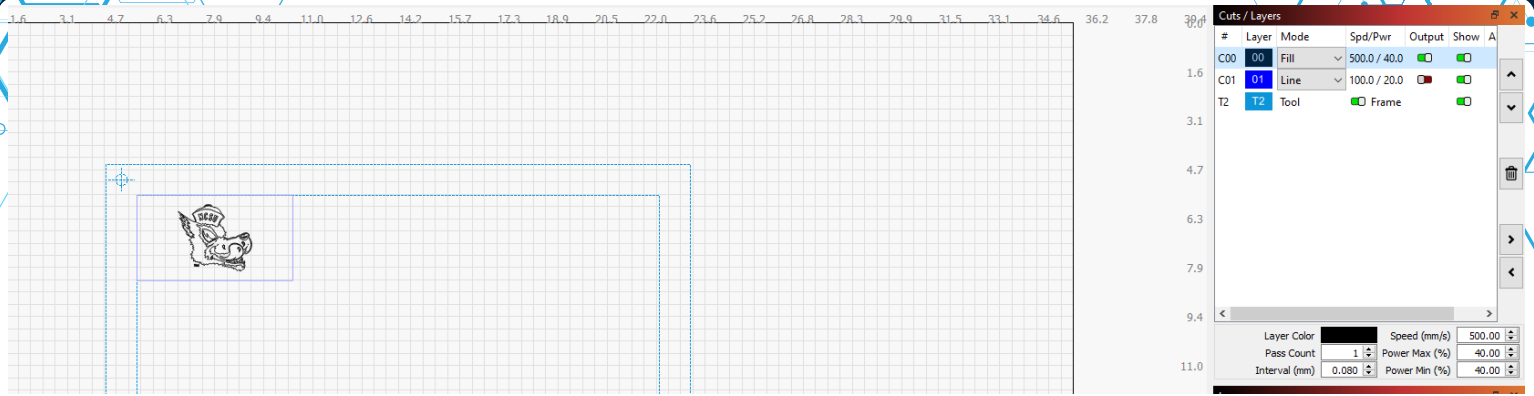
You can change mode in order to focus on

Line (Cut or Etch)

or

Fill (Rastering)





Currently, the Autofocus tool needs calibration, and this guide will be updated in the future. Utilize the manual acrylic tool to set the laser 6 mm from the material you're engraving. Note the thickness of the triangle is 6 mm.

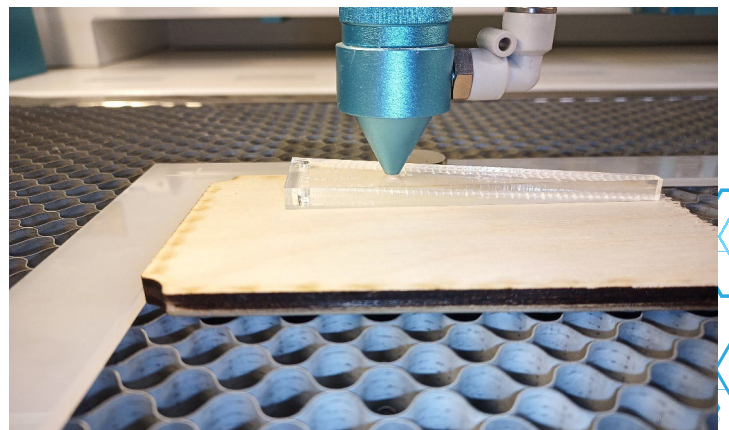
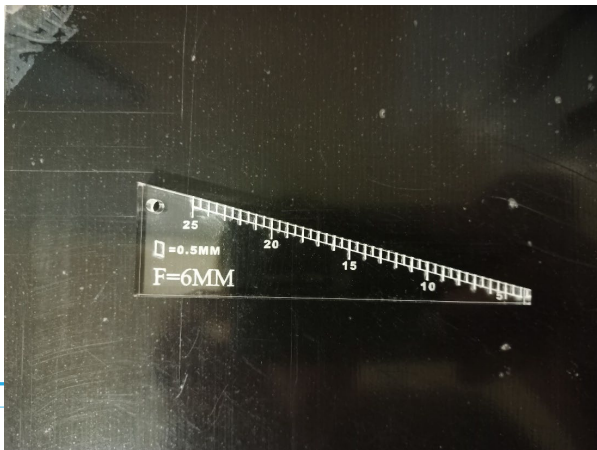
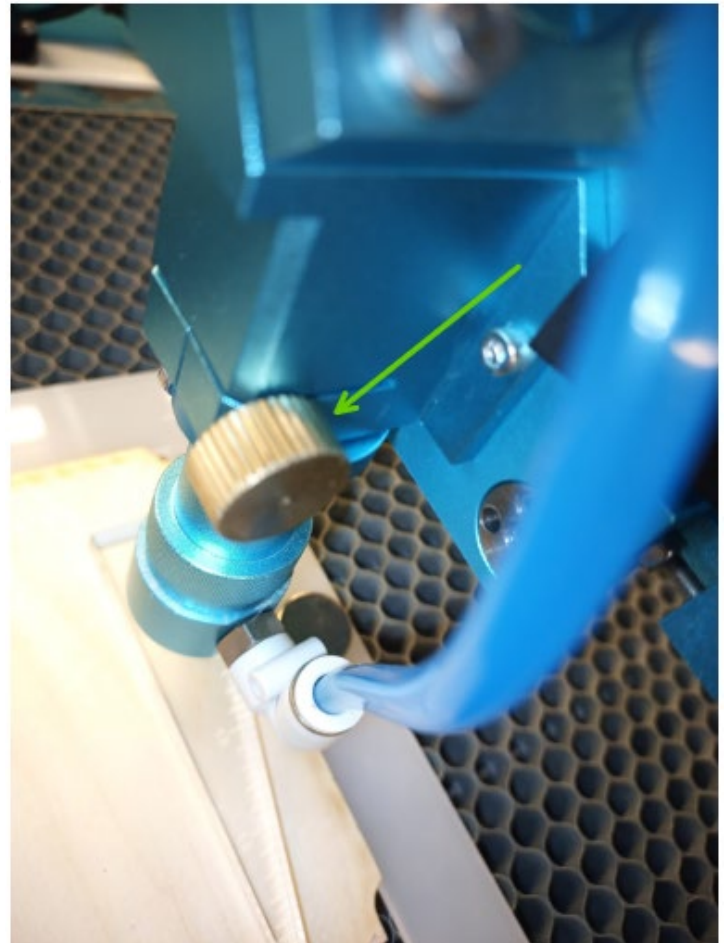
Ignore the ruler; it is for different focal lenses that have different focal distance requirements.

NCSU has ONE focal lens and 6 mm (THE THICKNESS OF THE TRIANGLE) is the recommended standard.

Bring the laser as close as possible, but do not utilize the buttons to touch the material.

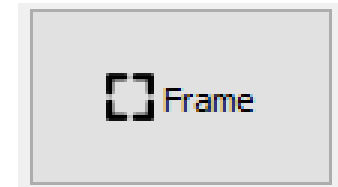
The laser SHOULD NOT slam into your project.

Utilize the golden screw as seen in the picture below to gently touch your project.

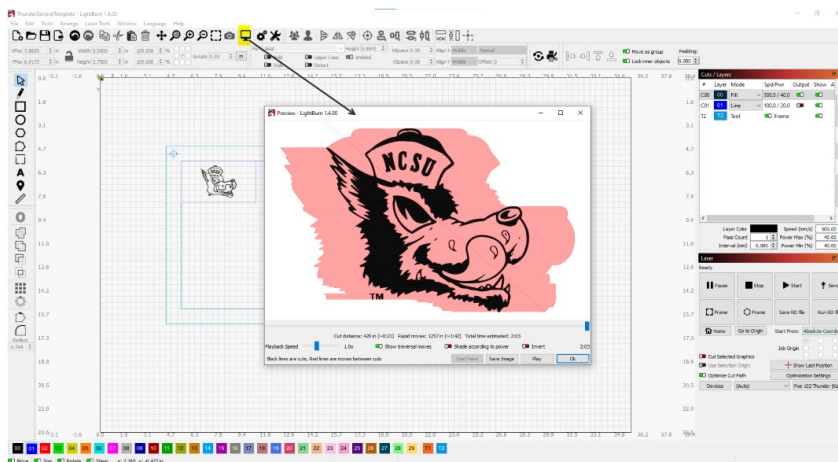


Remove the acrylic measuring tool AND PLACE IT BACK INTO THE YELLOW TOOLBOX FOR LASER TOOLS.

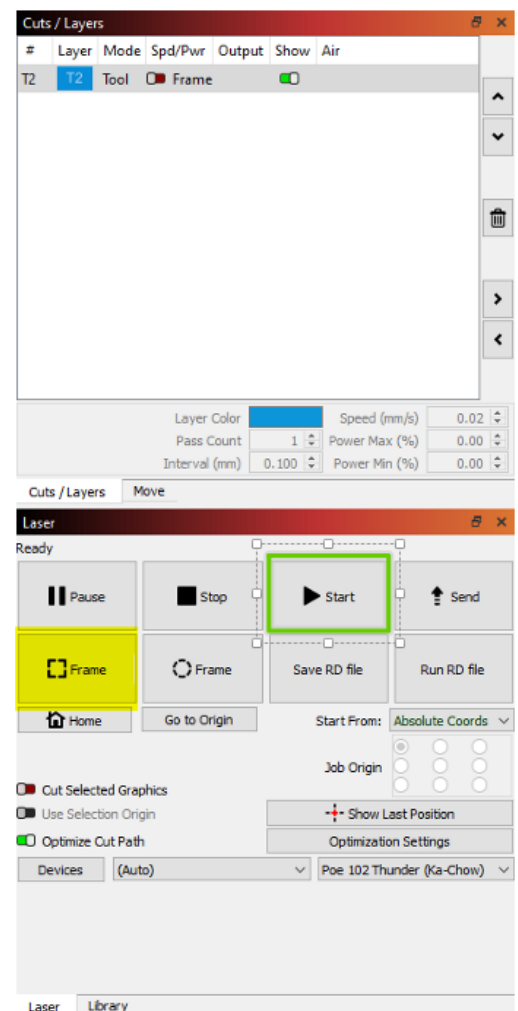
Click the Frame button to double-check that your laser is in the correct general location relative to your material.



Click the Preview button to ensure that your file will be lasered correctly.



After confirming that the laser is in the right location, hit the Start button to send it and create your project.



Admire your work.

